Telecommunication Systems
ICE 3109

Lecture 10
Digital Cellular Communication

The Cellular Concept

- A major breakthrough in solving the problem of
  - Spectral congestion
  - User capacity.
- It offered very high capacity in a limited spectrum allocation without any major technological changes.
- The cellular concept is a system-level idea
  - Replacing a single, high power transmitter (large cell) with many low power transmitters (small cells),
- Each base station is allocated a portion of the total number of channels available to the entire system
- Neighboring base stations are assigned different groups of channels so that the interference between base stations is minimized.

The Cellular Concept

- By systematically spacing base stations the available channels are distributed throughout the geographic region
- Provides additional radio capacity with no additional increase in radio spectrum.
- This fundamental principle is the foundation for all modern wireless communication systems
- Furthermore, the cellular concept allows every piece of subscriber equipment within a country or continent to be manufactured with the same set of channels so that any mobile may be used anywhere within the region.

Frequency Reuse

- Each base station is allocated a group of radio channels to be used within a small geographic area called a **cell.**
- Adjacent cells are assigned channel groups which contain completely different channels than neighboring cells.
- The base station antennas are designed to achieve the desired coverage within the particular cell.
- By limiting the coverage area of a cell, the same group of channels may be used to cover different cells
  - By keeping interference levels within tolerable limits.
- The design process of selecting and allocating channel groups for all of the cellular base stations within a system is called **frequency reuse** *or* **frequency planning**.

Frequency Reuse

- The actual radio coverage of a cell is known as the **footprint**
  - Determined from field measurements or propagation prediction models.
- The real footprint is amorphous in nature,
  - a regular cell shape is needed for systematic system design
- A circle to represent the coverage area of a base station
  - Adjacent circles cannot be overlaid upon a map without leaving gaps.
- To cover an entire region without overlap there are three sensible choices
  - A square, an equilateral triangle, and a hexagon.

Frequency Reuse

- For a given distance between the center of a polygon and its farthest perimeter points, the hexagon has the largest area of the three.
  - By using the hexagon geometry, the fewest number of cells can cover a geographic region, and the hexagon closely approximates a circular radiation pattern.

Frequency Reuse

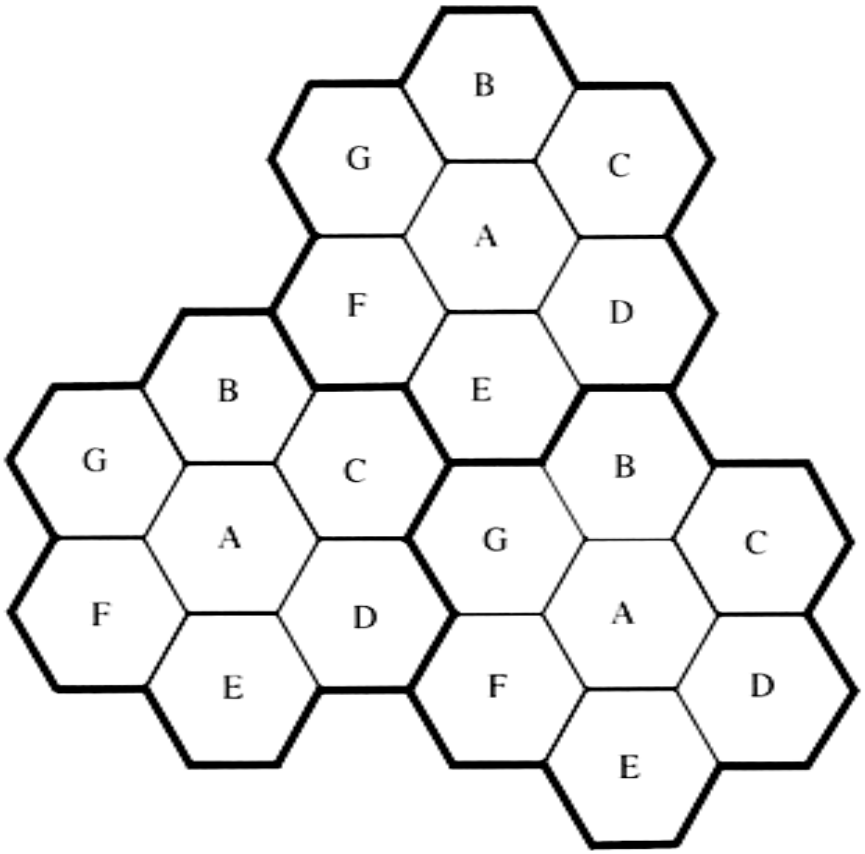


Figure 3.1 Illustration of the cellular frequency reuse concept. Cells with the same letter use the same set of frequencies. A cell cluster is outlined in bold and replicated over the coverage area. In this example, the cluster size, *N*, is equal to seven, and the frequency reuse factor is 1/7 since each cell contains one-seventh of the total number of available channels.

Frequency Reuse

- Consider a cellular system which has a total of *S* duplex channels available for use.
- If each cell is allocated a group of k channels *(k < S),* and if the *S* channels are divided among *N* cells the total number of available radio channels can be expressed as

$$S=Nk$$

- The *N* cells which collectively use the complete set of available frequencies is called a **cluster**.
- If a cluster is replicated *M* times within the system, the total number of duplex channels, *C*, can be used as a measure of capacity and is given by

$$C=MkN=MS$$

Co-Channel: The channels which are replicated in different cluster is called Co channel.

Frequency Reuse

- The factor *N* is called the cluster size
  - Typically equal to 4, 7, or 12.
- *N* is reduced while the cell size is kept constant,
  - More clusters are required to cover a given area,
  - More capacity (a larger value of *C*) is achieved.
- The value for *N* is a function of how much interference a mobile or base station can tolerate.
- From a design viewpoint, the smallest possible value of N is desirable to maximize capacity over a given coverage area.
- The **frequency reuse factor** of a cellular system is given by *1/N*, since each cell within a cluster is only assigned *1/N* of the total available channels in the system.

Frequency Reuse

- The geometry of hexagons is such that the number of cells per cluster, *N, can only have values which satisfy Equation*

$$N=i^2+ij+j^2$$

  - where *i* and *j* are non-negative integers.
- To find the nearest co-channel neighbors of a particular cell, one must do the following:
  1. Move *i* cells along any chain of hexagons
  2. Turn 60 degrees counter-clockwise and move *j* cells.

Frequency Reuse

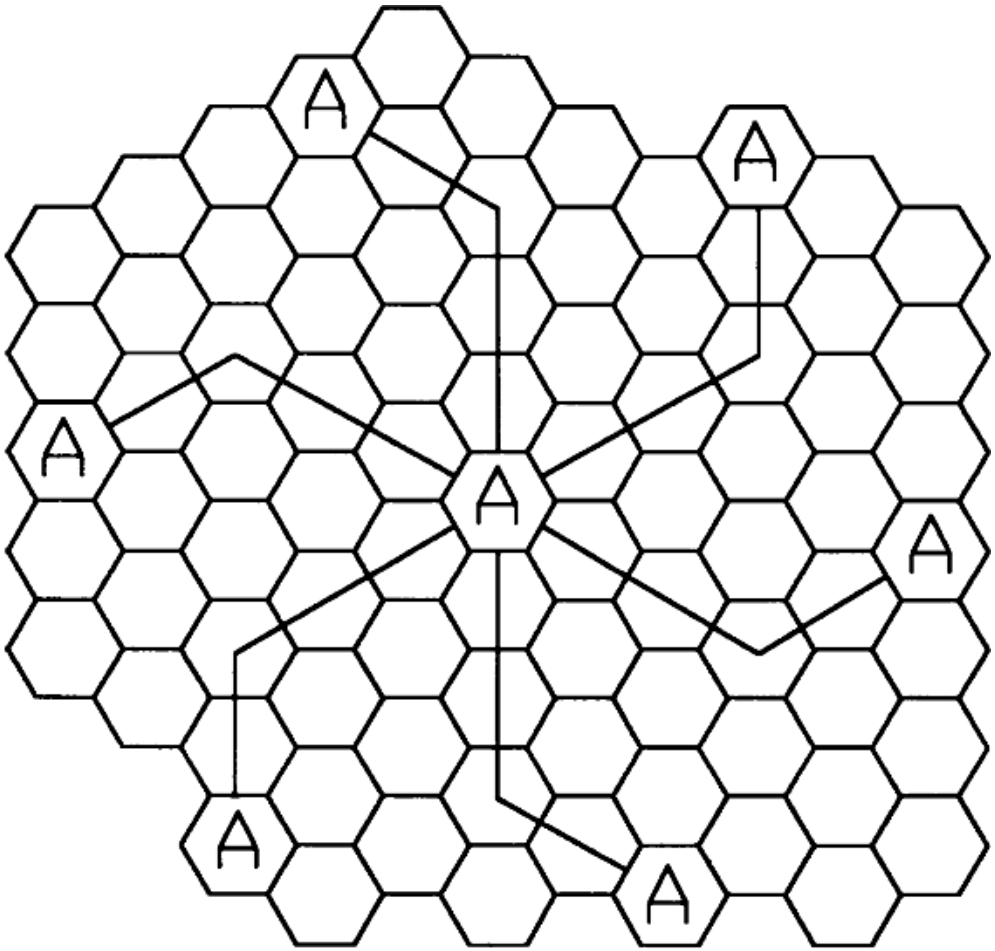


Figure 3.2 Method of locating co-channel cells in a cellular system. In this example, $N = 19$ (i.e., $I = 3$, $j = 2$). (Adapted from [Oet83] © IEEE.)

Frequency Reuse

- Example
  - If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 25 kHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses
    a) four-cell reuse,
    b) seven-cell reuse, and
    c) 12-cell reuse.
  - If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems.

Frequency Reuse

- Solution
  - Given:
    - Total bandwidth = 33 MHz
    - Channel bandwidth = 25 kHz × 2 simplex channels = 50 kHz/duplex channel
  - Total available channels = 33,000/50 = 660 channels

a) For N = 4,

total number of channels available per cell = 660/4 ≈ 165 channels.

b) For N = 7,

total number of channels available per cell = 660/7 ≈ 95 channels.

c) For N = 12,

total number of channels available per cell = 660/12 ≈ 55 channels.

Frequency Reuse

- Solution (Cont.)
  - A 1 MHz spectrum for control channels implies that there are 1000/50 = 20 control channels out of the 660 channels available.

a) For N = 4, we can have five control channels and 160 voice channels per cell.

In practice, however, each cell only needs a single control channel. Thus, one control channel and 160 voice channels would be assigned to each cell.

b) For N = 7, four cells with three control channels and 92 voice channels, two cells with three control channels and 90 voice channels, and one cell with two control channels and 92 voice channels could be allocated.

In practice, however, each cell would have one control channel, four cells would have 91 voice channels, and three cells would have 92 voice channels.

c) For N = 12, we can have eight cells with two control channels and 53 voice channels, and four cells with one control channel and 54 voice channels each.

In an actual system, each cell would have one control channel, eight cells would have 53 voice channels, and four cells would have 54 voice channels.

Channel Assignment Strategies

- Goal
  - increasing capacity
  - minimizing interference
- Classification
  - *Fixed*
  - *Dynamic*

Channel Assignment Strategies

- Fixed channel assignment strategy
  - Each cell is allocated a predetermined set of voice channels.
  - Any call attempt within the cell can only be served by the unused channels
  - If all the channels in that cell are occupied, the call is *blocked .*
    - Subscriber does not receive service.

Channel Assignment Strategies

- Variations of the fixed assignment strategy
  - The *borrowing strategy*
    - *A cell is allowed to borrow channels from a neighboring* cell if all of its own channels are already occupied.
    - The mobile switching center (MSC) supervises such borrowing procedures
    - Ensures that the borrowing of a channel does not disrupt or interfere with any of the calls in progress in the donor cell.

Channel Assignment Strategies

- Dynamic channel assignment strategy,
  - Voice channels are not allocated to different cells permanently.
    - Each time a call request is made, the serving base station requests a channel from the MSC.
  - The switch then allocates a channel to the requested cell by considering
    - Likelihood of future blocking within the cell,
    - The frequency of use of the candidate channel,
    - The reuse distance of the channel, and
    - Other cost functions.

Handoff Strategies

- A mobile moves into a different cell while a conversation is in progress,
  - The MSC automatically transfers the call to a new channel belonging to the new base station.
  - This handoff operation
    - Identifying a new base station,
    - Requires that the voice and control signals be allocated to channels associated with the new base station.

It is the process of transferring an active call from one cell to another cell when the mobile unit moves from one cell to another cell without disconnecting the call.

Handoff Strategies

- Processing handoffs
  - An important task in any cellular radio system.
  - Prioritize handoff requests over call initiation requests when allocating unused channels
  - Handoffs must be performed successfully and as infrequently as possible, and be imperceptible to the users.
  - In order to meet these requirements, system designers must specify an optimum signal level at which to initiate a handoff.

Handoff Strategies

- Processing handoffs (cont.)
  - A particular signal level is specified as the minimum usable signal for acceptable voice quality at the base station receiver
    - Normally taken as between –90 dBm and –100 dBm
  - A slightly stronger signal level is used as a threshold at which a handoff is made.
  - If Δ is too large, unnecessary handoffs may occur
  - If Δ is too small, there may be insufficient time to complete a handoff.

Handoff Strategies

- Deciding when to handoff
  - The drop in the measured signal level is not due to momentary.
  - To ensure this, the base station monitors the signal level for a certain period of time before a handoff is initiated.
  - This running average measurement of signal strength should be optimized so that unnecessary handoffs are avoided.

Handoff Strategies

- 2G systems handoff decisions
  - Mobile assisted.
  - In mobile assisted handoff (MAHO), every mobile station measures the received power from surrounding base stations
  - Continually reports the results to the serving base station.
    - A handoff is initiated when the power received from the base station of a neighboring cell begins to exceed the power received from the current base station by a certain level or for a certain period of time.

Handoff Strategies

- Prioritizing Handoffs
  - The *guard channel concept,*
    - *A fraction* of the total available channels in a cell is reserved exclusively for handoff requests
    - Disadvantage of reducing the total carried traffic
  - Queuing of handoff requests
    - Possible because a finite time interval when signal level drops below the handoff threshold.
    - Decrease the probability of forced termination of a call
    - There is a tradeoff between the decrease in probability of forced termination and total carried traffic.

Interference and System Capacity

- Interference is the major limiting factor in the performance of cellular radio systems.
- Sources of interference
  - Another mobile in the same cell,
  - A call in progress in a neighboring cell,
  - Other base stations operating in the same frequency band,
  - Any noncellular system which inadvertently leaks energy into the cellular frequency band.

Interference and System Capacity

- Impacts of interference
  - Interference on voice channels causes cross talk,
    - The subscriber hears interference in the background due to an undesired transmission.
  - On control channels, interference leads to missed and blocked calls
    - Errors in the digital signaling.

Interference and System Capacity

- Impacts of interference
  - Interference is more severe in urban areas,
    - Greater RF noise floor
    - The large number of base stations and mobiles.
  - Interference has been recognized as a major bottleneck in increasing capacity
  - It is often responsible for dropped calls.
- The two major types of cellular interference
  - *Co-channel interference*
  - *Adjacent channel interference.*

Interference and System Capacity

- Co-channel Interference
  - Frequency reuse
    - Several cells that use the same set of frequencies.
  - These cells are called co-channel cells
  - The interference between signals from these cells is called co-channel interference.
  - Unlike thermal noise
    - Thermal noise can be overcome by increasing the SNR co-channel interference cannot be combated by imply increasing the carrier power of a transmitter.

Interference and System Capacity

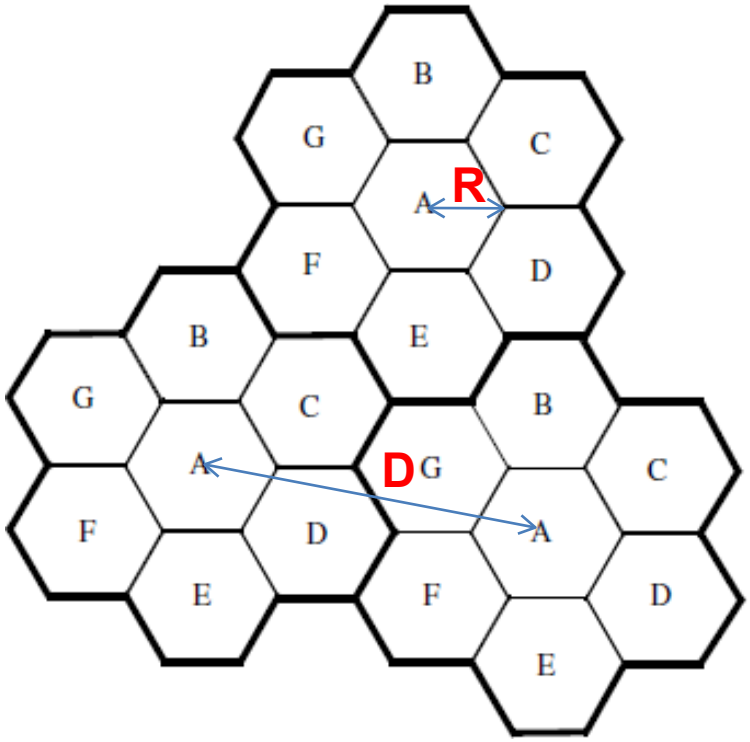
- Co-channel Interference
  - Increase in carrier transmit power increases the interference to neighboring co-channel cells.
  - To reduce co-channel interference, co-channel cells must be physically separated by a minimum distance to provide sufficient isolation due to propagation.

Interference and System Capacity

- Co-channel Interference
  - The co-channel interference ratio is independent of the transmitted power
  - It is a function of the radius of the cell (*R*) and the distance between centers of the nearest co-channel cells (*D).*
  - By increasing the ratio of *D/R,* the spatial separation between co-channel cells is increased.
  - Thus, interference is reduced from improved isolation of RF energy from the co-channel cell.

Interference and System Capacity

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Interference and System Capacity

- Co-channel Interference
  - The parameter Q, called the co-channel reuse ratio. For a hexagonal geometry

$$Q = \frac{D}{R} = \sqrt{3N}$$

  - A small value of Q provides larger capacity since the cluster size N is small
  - *A* large value of Q improves the transmission quality
  - A trade-off must be made between these two objectives in actual cellular design.

Interference and System Capacity

- Channel Planning for Wireless Systems
  - Assigning the appropriate radio channels to each base station is much more difficult in practice than in theory.
  - Equation of SIR is a valuable rule of thumb for determining the appropriate frequency reuse ratio
  - The wireless engineer must deal with the real-world difficulties of radio propagation and imperfect coverage regions of each cell.
  - Cellular systems, in practice, seldom obey the homogenous propagation path loss assumption of SIR equation .

Interference and System Capacity

- Channel Planning for Wireless Systems
  - Generally, the available mobile radio spectrum is divided into channels
  - *These channels generally are* made up of
    - control channels - vital for initiating, requesting, or paging a call
    - voice channels- dedicated to carrying revenue-generating traffic.
  - Typically, about 5% is devoted to control channels,
    - Very brief and bursty in nature,
  - Remaining 95% of the spectrum is dedicated to voice channels.

Interference and System Capacity

- Channel Planning for Wireless Systems
  - In practical systems, a distinction between voice and control channels
    - control channels are generally not allowed to be used as voice channels, and vice versa.
  - Control channels are vital in the successful launch of any call,
    - The frequency reuse strategy applied to control channels is different and generally more conservative (e.g., is afforded greater S/I protection) than for the voice channels.

Interference and System Capacity

- Channel Planning for Wireless Systems
  - In CDMA systems the cluster size is N = 1, and frequency planning is not nearly as difficult as for TDMA or first generation cellular systems .
  - In CDMA, a single 1.25 MHz radio channel carries the simultaneous transmissions of the single control channel with up to 64 simultaneous voice channels.
  - Time varying coverage region which varies depending on the instantaneous number of users on the CDMA radio channel

Interference and System Capacity

- Adjacent Channel Interference
  - Interference resulting from signals which are adjacent in frequency to the desired signal is called *adjacent channel interference.*
  - *Adjacent channel interference results from imperfect receiver filters*
    - Allow nearby frequencies to leak into the passband.
  - The problem can be particularly serious if an adjacent channel user is transmitting in very close range to a subscriber's receiver.Then problem arise and this is called near far effect.
  - This is referred to as the *near–far effect*

Interference and System Capacity

- Adjacent Channel Interference
  - The near–far effect
    - A nearby transmitter (which may or may not be of the same type as that used by the cellular system) captures the receiver of the subscriber.
    - Alternatively, the near–far effect occurs when a mobile close to a base station transmits on a channel close to one being used by a weak mobile.

Interference and System Capacity

- Adjacent Channel Interference
  - Adjacent channel interference can be minimized through careful filtering and channel assignments.
  - Since each cell is given only a fraction of the available channels, a cell need not be assigned channels which are all adjacent in frequency.
  - By keeping the frequency separation between each channel in a given cell as large as possible, the adjacent channel interference may be reduced considerably.
  - Some channel allocation schemes also prevent a secondary source of adjacent channel interference by avoiding the use of adjacent channels in neighboring cell sites.

Interference and System Capacity

- Power Control for Reducing Interference
  - The power levels transmitted by every subscriber unit are under constant control by the serving base stations.
  - This is done to ensure that each mobile transmits the smallest power necessary to maintain a good quality link
  - Power control not only helps prolong battery life for the subscriber unit, but also dramatically reduces the reverse channel *S/I in the system*